algorithms work; in chapter 14 we did it manually and took it for granted, but what about an automated approach to embed these data structure in a Euclidean space, and in particular in the 2D plane?

This is not always needed for all graphs' applications, nor it is always possible; there are, however, many applications where the way we lay a graph's vertices and edges on a surface are crucial. Take, for instance, printed circuits boards (*PCB*) design, shown in figure 15.1: the way electronic components (vertices) and conductive tracks (edges) are positioned over the board is crucial not just to the good functioning of the circuit, but also to optimize the manufacturing process and reduce the amount of copper used, and the overall costs.

In this chapter, we are going to gradually introduce the main concepts about graph embeddings, focusing on 2D planes, planar graphs, and how to minimize the intersections of the edges when a graph is not planar.

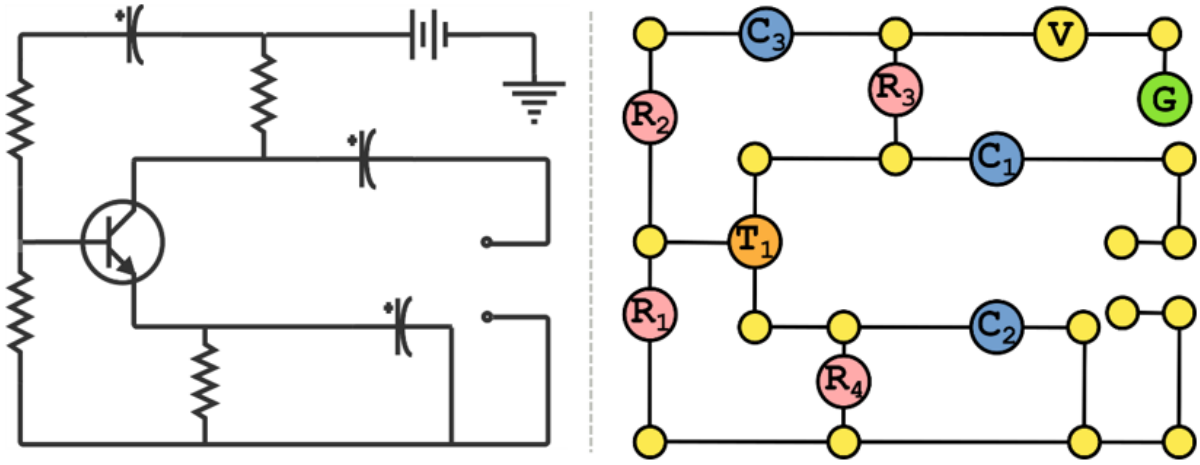


Figure 15.1 An example of an electronic circuit, and a graph embedding from which the circuit's layout